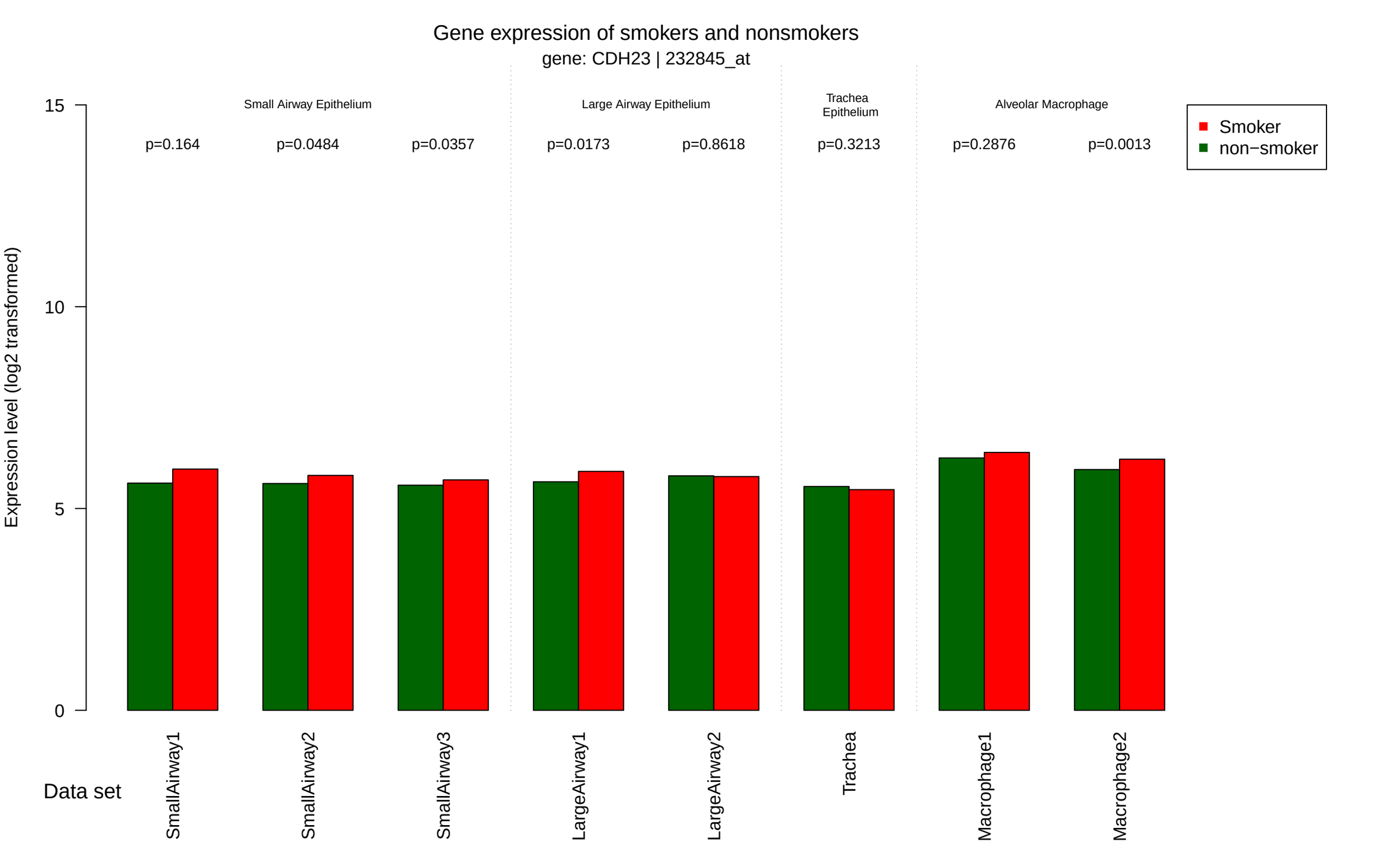
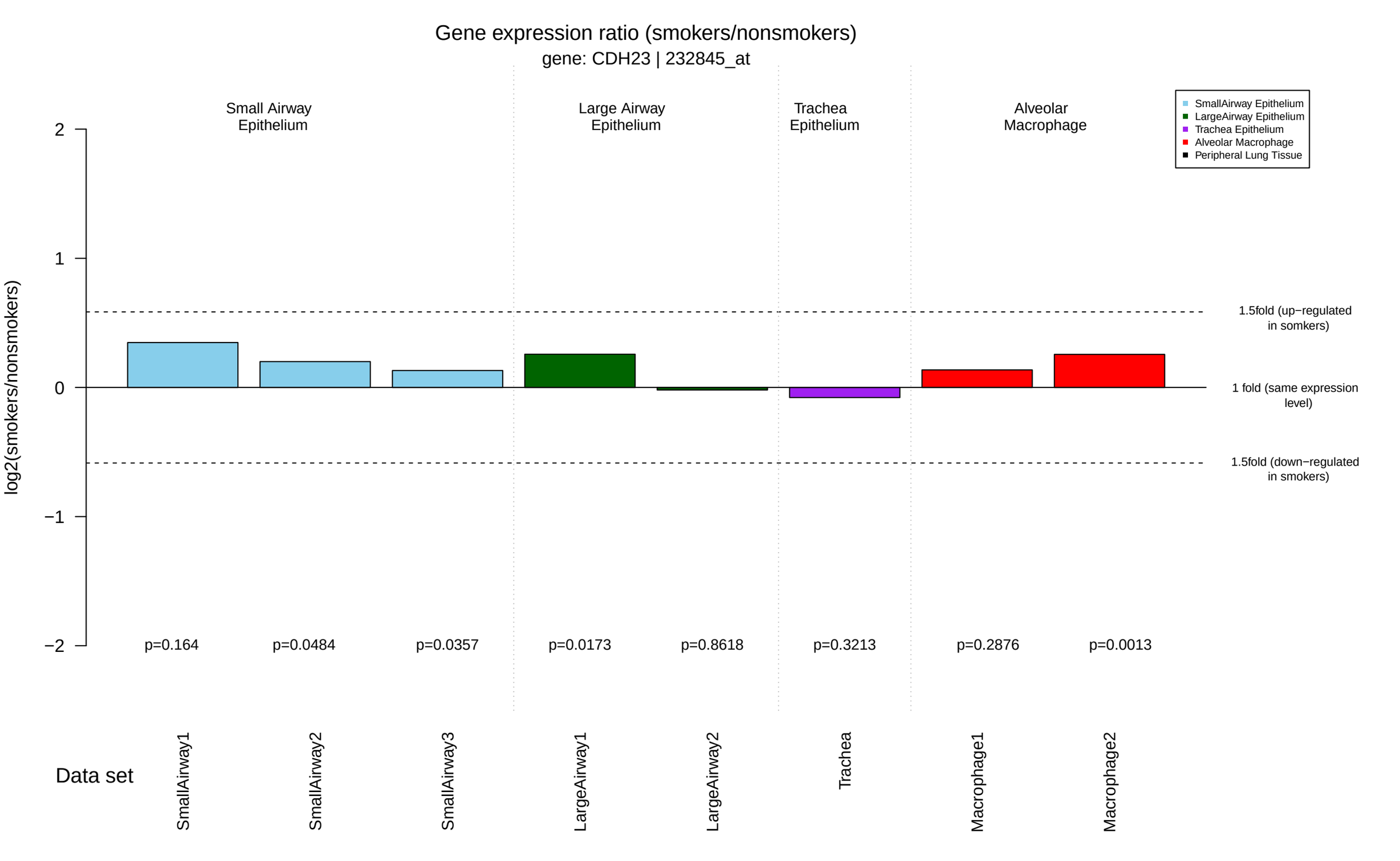
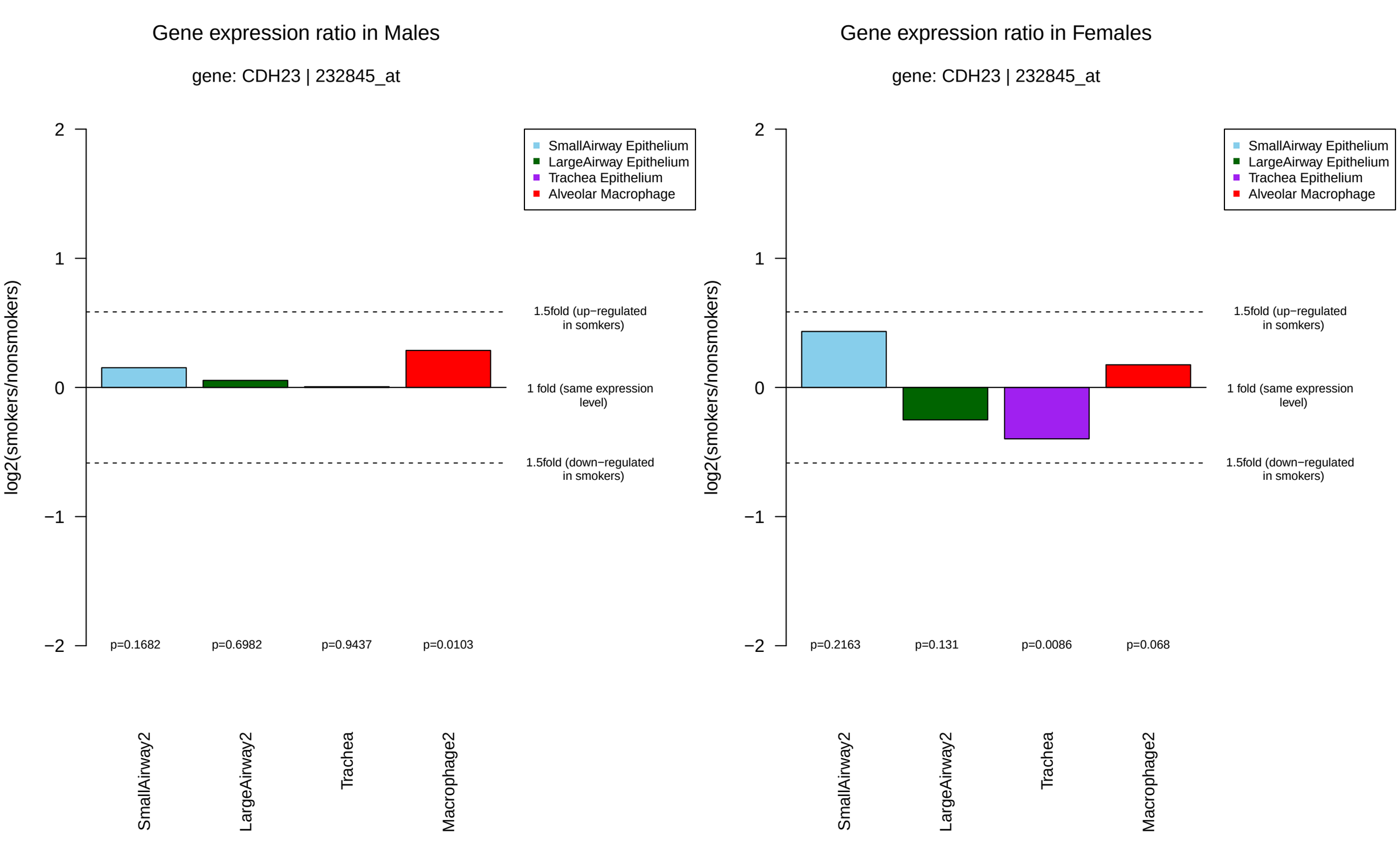
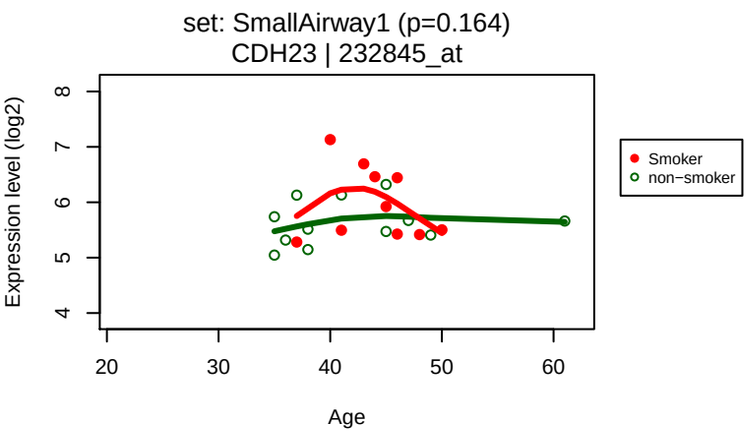
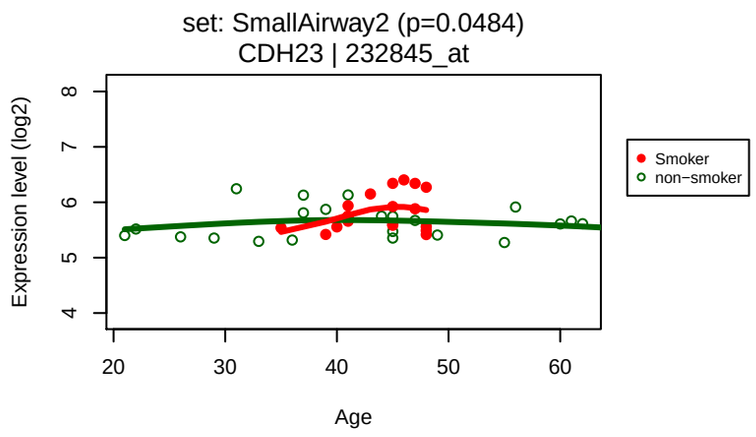
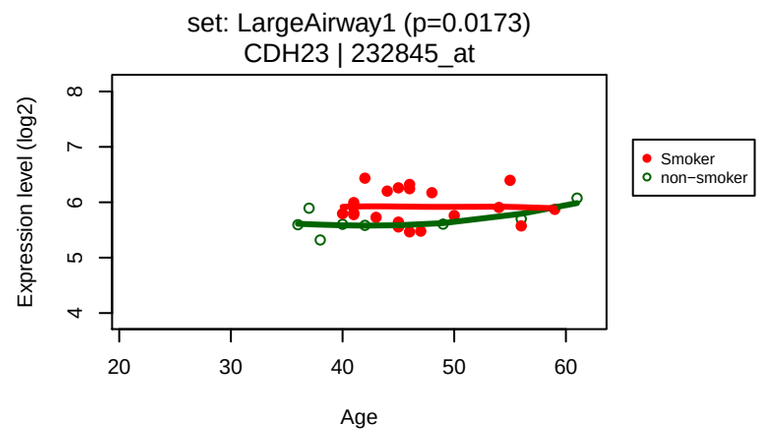

Gene expression of smokers and nonsmokers
gene: CDH23 | 232845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.164
p=0.0484
p=0.0357
p=0.0173
p=0.8618
p=0.3213
p=0.2876
p=0.0013
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

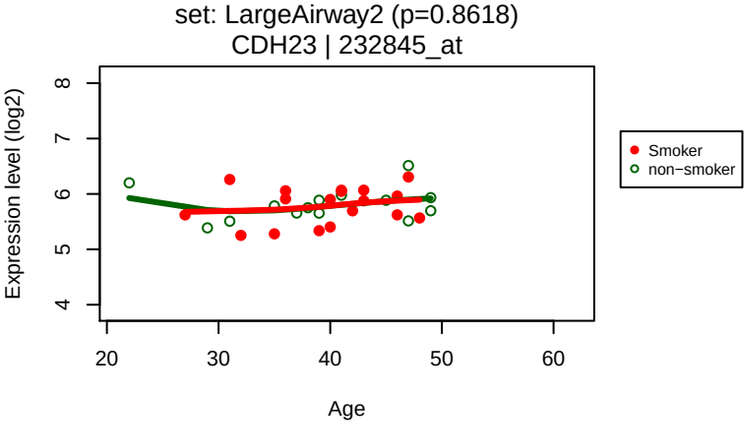

Gene expression ratio (smokers/nonsmokers)
gene: CDH23 | 232845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.164
p=0.0484
p=0.0357
p=0.0173
p=0.8618
p=0.3213
p=0.2876
p=0.0013
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

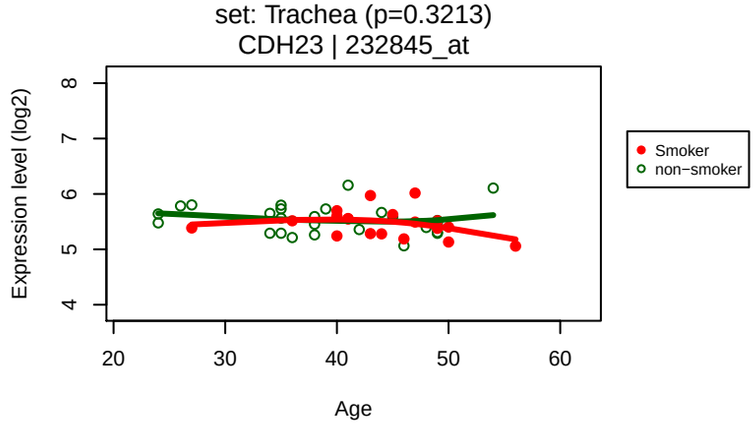

Gene expression ratio in Males
gene: CDH23 | 232845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1682
p=0.6982
p=0.9437
p=0.0103
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDH23 | 232845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2163
p=0.131
p=0.0086
p=0.068
SmallAirway2
LargeAirway2
Trachea
Macrophage2

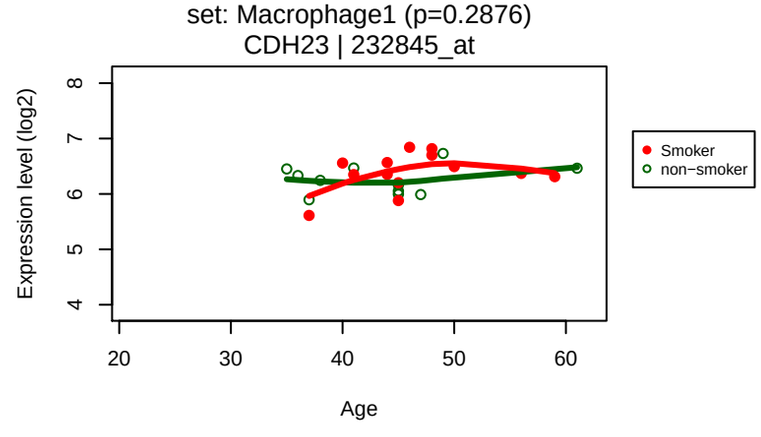

set: SmallAirway1 (p=0.164)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker

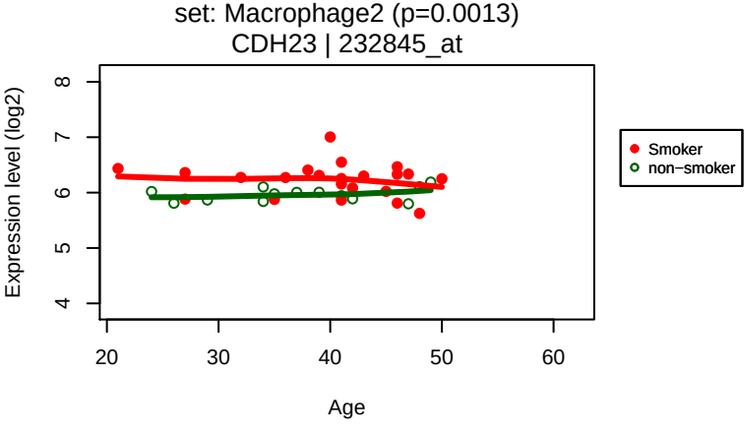

set: SmallAirway2 (p=0.0484)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker

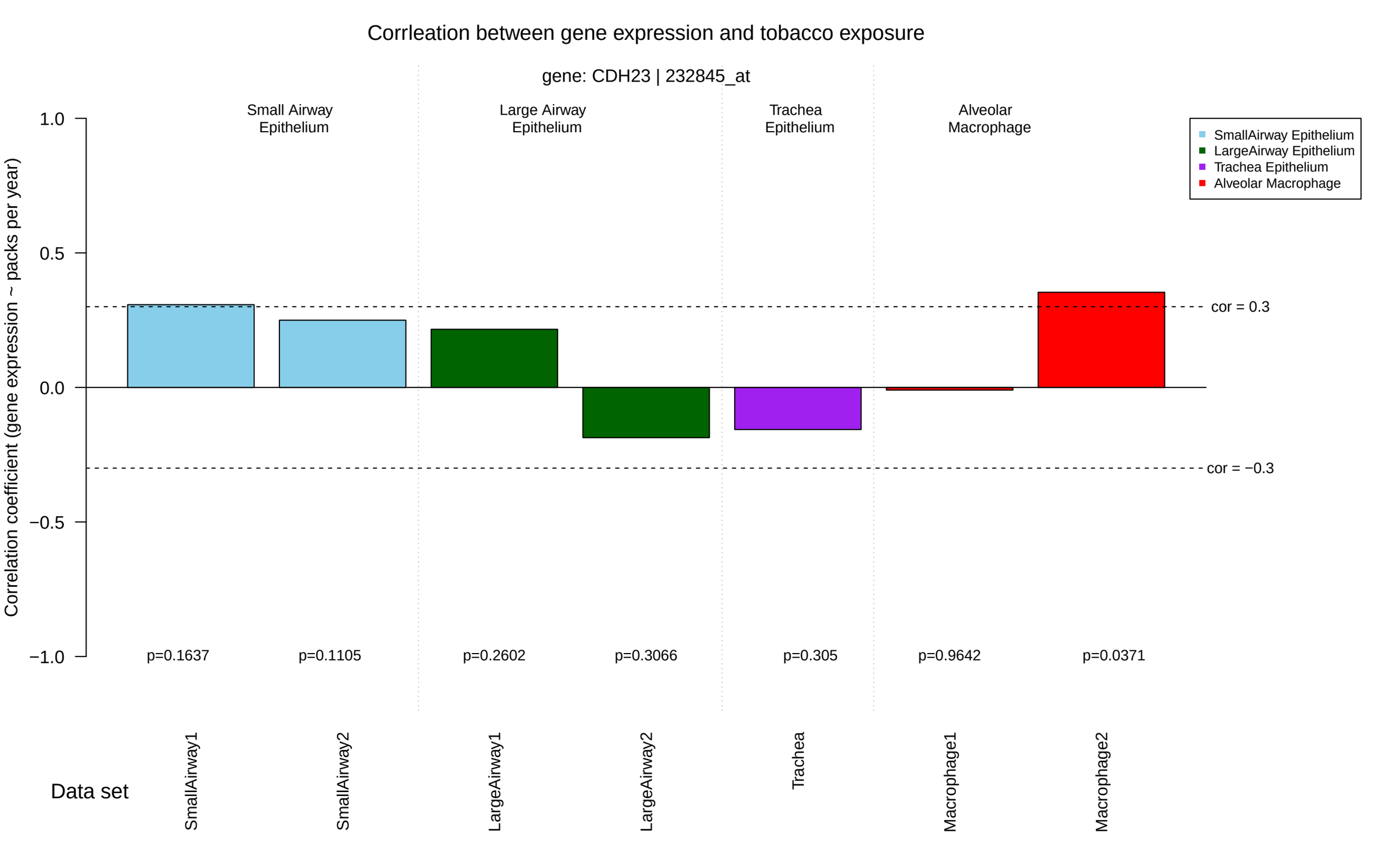

set: LargeAirway1 (p=0.0173)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8618)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3213)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2876)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0013)
CDH23 | 232845_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDH23 | 232845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1637
p=0.1105
p=0.2602
p=0.3066
p=0.305
p=0.9642
p=0.0371
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2